

Review Immnn

정영두

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- Deep neural networks (DNNs) typically implicitly assume that observed responses are statistically independent.
- Real-world data, however, “include spatial, temporal and clustering structures.”

In a typical LMM setting $y \in \mathbb{R}^n$ is a dependent variable modeled by X and Z , which are $n \times p$ and $n \times q$ model matrices respectively

$$y = X\beta + Zb + \varepsilon$$

1. “We propose to use the mixed-models framework to handle correlated data in DNNs.”

$$y = f(X) + g(Z)b + \varepsilon$$

2. “The key is using the Gaussian negative log-likelihood (NLL) as a natural loss function that is minimized with DNN machinery including stochastic gradient descent (SGD).”
3. Mini-batch trick: “We calculate the inverse of the $m \times m$ sub-matrix $V \dots$ This ‘inversion in parts’ is the key element behind LMMNN’s scalability.”

Multiple categorical features

논문에 있는 실제 데이터 중에서 InstEval를 사용

InstEval Description: 73K students 1-5 evaluations of professors from ETH Zurich

Dataset	n	K	p	categorical	q	y
InstEval	73 K	3	3	student teacher department	2.9 K 1.1 K 14	Teacher ratings (1-5)

`data.head()`

	students	instructors	studage	lectage	service	departments	ratings
0	0	524	2	2	0	1	5
1	0	559	2	1	1	5	2
2	0	831	2	2	0	1	5
3	0	1067	2	2	1	2	3
4	1	61	2	1	0	4	2

1. 연속형 변수(고정효과)

`add_layers_functional(X_input, n_neurons, dropout, activation)` 을 통해

$3 \rightarrow 10 \rightarrow 3 \rightarrow 1$ 구조의 MLP(Multi-Layer Perceptron)를 거쳐 $\hat{y}$ 을 만들

2. 범주형 변수(랜덤효과)

▪ 선형인 경우

Z를 One-hot 또는 Identity(그대로) 등을 반환해서 NLL(Negative Log-Likelihood)에 입력

예) $y \sim N(f(X), g(Z)Dg(Z)^\top + \sigma_e^2 I)$

▪ 비선형인 경우

Z를 임베딩, MLP로 바꾸는 변환의 경우는 $g(Z)$ 변환해서 NLL(Negative Log-Likelihood)에 입력

예) $y | b \sim N(f(X) + g(Z)b, \sigma_e^2 I)$

3. NLL의 최적값 구하기

$\sigma_e^2, \sigma_{b_k}^2$ 등 구해야 할 모수를 Adam 최적화 + EarlyStopping(개선이 이루어지지 않으면 멈춤)로 찾기

4. BLUP(Best Linear Unbiased Predictor)계산

$$\hat{b} = D(\hat{\psi})\hat{g}(Z_{tr})' V(\hat{g}, \hat{\theta})^{-1} (y_{tr} - \hat{f}(X_{tr}))$$

5. 예측값 구하기

`model.predict(...)`에서 반환하는 고정효과 $f(X)$

선형 또는 비선형으로 구해진 $g(Z)$ 와 $\hat{b}$ 를 곱한것을

둘이 더해서 y 의 예측값을 구한다.

$$\hat{y} = \hat{f}(X) + \hat{g}(Z)\hat{b}$$

6. MSE를 구한다.

exp_type	mean_mse	std_mse
embed	1.5559	0.1148
ignore	1.7774	0.0155
lmm	1.4757	0.0138